

An Explainable Cleanlab-Guided Uncertainty-Based Active Learning Framework for Data-Efficient Parkinson’s Disease Classification

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Abstract. Parkinson’s Disease (PD) is a progressive neurodegenerative disorder whose early diagnosis is challenging due to dependence on expert-driven clinical assessment and the limited availability of labeled medical data. To address this issue, this study proposes a Cleanlab-Guided Uncertainty-Based Active Learning (AL) framework for voice-based PD detection using high-dimensional acoustic features. The proposed approach iteratively selects the most informative and uncertain samples from an unlabeled pool, significantly reducing annotation cost while maintaining strong classification performance. Experimental results on a benchmark PD voice dataset demonstrate that the model achieves 94.0% accuracy, 94.0% F1-Score, and 88.1% MCC, with rapid convergence within eight AL rounds. A comprehensive ablation study is conducted to evaluate the contribution of different feature groups, revealing that TQWT and MFCC features are the most critical, where their removal leads to notable performance degradation, particularly a significant drop in MCC. Furthermore, a comparative analysis with state-of-the-art Semi-Supervised Learning models demonstrates that the proposed framework consistently outperforms existing methods. To enhance interpretability, SHAP and LIME are employed for global and local explanations, respectively, providing insights into feature importance and model decision behavior. Overall, the integration of uncertainty-driven AL with explainable AI offers a scalable, cost-effective, and reliable framework for early PD detection in data-scarce clinical environments.

Keywords: Parkinson’s Disease · Active Learning · Cleanlab · Uncertainty Sampling · Voice-Based Diagnosis · Explainable AI.

1 Introduction

Parkinson’s disease (PD) is a progressive neurodegenerative disorder characterized by the gradual loss of dopaminergic neurons in the substantia nigra of the midbrain, leading to significant motor dysfunction and impaired coordination [1, 2]. As the second most common neurodegenerative disease worldwide, PD is characterized by symptoms such as bradykinesia, tremor, rigidity, and postural instability, often accompanied by cognitive

and emotional impairments in later stages. Conventional PD diagnosis is based on the clinical neurology assessment and general rating scale, such as the Unified Parkinson Disease Rating Scale (UPDRS) [3, 4]. Such techniques have well-trained clinicians, and in most cases, they are based on subjective evaluation, which it is difficult to diagnose early. Here, automated diagnostic technologies with data have become increasingly popular as tools to assist physicians. Recent advancements in Machine Learning (ML) and Artificial Intelligence (AI) have enabled automated diagnostic systems that analyze biological data for disease detection [5, 6]. Speech analysis, particularly voice analysis, is a promising non-invasive method for PD detection, as speech disorders are among the earliest symptoms of PD. Early and objective diagnosis is possible using AI-based systems that are able to identify subtle effects in high-dimensional audio features.

This paper suggests a Cleanlab-Guided Uncertainty-Based Active Learning voice-based PD recognition framework to solve these problems. In contrast to conventional forms of supervised and Semi-Supervised Learning (SSL), Active Learning (AL) is an iterative approach that uses uncertainties and informative samples to select the most informative samples in an unlabeled data pool to be annotated. The proposed method reduces the redundancy of labeling, reduces the costs of annotation, and improves model convergence by only focusing on samples that are most useful in refining the decision boundary. This customized sampling technique can be done faster compared to random selection to promote performance. Moreover, the proposed system demonstrates that a relatively small amount of labeled data can be used to achieve high diagnostic accuracy, implying that one may not need substantial annotation. It is compared to its performance in various AL rounds. The major contributions of this study are summarized below.

- This work introduces a novel integration of Cleanlab-driven uncertainty estimation with AL for voice-based PD classification, reducing dependence on fully labeled datasets.
- A selective uncertainty sampling strategy is used to identify the most informative samples, thereby minimizing labeling cost while maintaining high diagnostic performance.
- Shapley Additive Explanations (SHAP) and Local Interpretable Model-agnostic Explanations (LIME) are used for global and local interpretability to identify key features for PD classification, supported by a comprehensive ablation study to assess feature group contributions, and a comparison with state-of-the-art SSL models.

2 Related Work

Recent research on PD detection has placed an increasing focus on data-efficient learning approaches because of the expensive nature and lack of availability of labeled medical datasets. Traditional supervised methods achieve strong performance but require fully labeled data, while SSL approaches partially reduce labeling dependence without incorporating selective sampling mechanisms. AL has become the most promising solution by querying the most informative samples rather than all of them. Yue et al. [7] up-sampled and applied AL to find that the accuracy increased from 44.3% with annotated 25% of samples; but, they did not focus on developing a standardized voice-based PD diagnosis. Moreno-Acevedo et al. [8] applied AL to multilingual PD speech samples, which showed

better cross-language generalization and required more data for labeling; however, their experiment consisted of a few participants. In parallel, Bao et al. [9] proposed an SSL competitive learning model with an 83.8% accuracy and 67.5% MCC; although the analysis is conducted on a small sample dataset. Conventional supervised methods by Sakar et al. [10] achieved a high accuracy of 89% and an F1-Score of 84%, though they need complete labeling of the dataset and do not use selective labeling. Adnan et al. [11] proposed an SSL deep neural network framework for voice-based PD detection, achieving 85.7% accuracy and 81.08% precision. Their model depended strongly on labeled data quality and lacked dataset diversity. More recently, Hasan et al. [12] suggested a small-sample learning method in gait classification of PD, and a limiting factor to generalizability is a small dataset, as well as controlled environments, and more complex data-efficient. In addition to the PD-related research, the latest trends in research on AL also point to the significance of effective data annotation methods. A hybrid framework based on AL and data programming, suggested by Nashaat et al. [13], and it is demonstrated to be able to attain similar accuracy as AL at much lower labeling costs on large-scale industrial datasets. More recently, Wang et al. [14] introduced an AL framework that has label quality control, which not only selects informative samples, but also detects and removes mislabeled data, further enhancing model robustness and efficiency.

With the advancement of research on AL, more complicated sampling plans have been proposed. Li et al. [15] introduced a list-wise learning-to-rank strategy that enhances sample selection by focusing on globally informative instances instead of just using uncertainty. Also, little has been undertaken to integrate uncertainty-based sampling to detect PD with explainable AI. This work aims to fill these gaps by suggesting an approach that can positively influence the quality of labels and select the most instructive cases to be annotated. Combining explainable AI methods with uncertainty-sensitive sampling, the proposed system can minimize the need to label data, enhance model accuracy, and promote transparency in decision-making, which is why it is perfectly suitable to be applied in clinical settings.

3 Methodology

This section outlines the systematic workflow of the proposed PD classification architecture. Voice-based acoustic features are collected, preprocessed, and used to train the model, which is then evaluated using multiple classification metrics.

3.1 Dataset Description

The dataset in this study is acquired in the UCI Machine Learning Repository and includes a benchmark of 744 acoustic features created out of voice recordings of PD patients and healthy controls [16]. The data consists of voice recordings of 188 PD patients (107 men and 81 women) and 64 healthy people (23 men and 41 women). Voice samples are all recorded at a frequency of 44.1 kHz. Following the clinical assessment, participants are required to make some repetitions of the sustained vowel /a/, and three audio recordings are made of each participant. Typically, Mel-Frequency Cepstral Coefficients (MFCCs), Wavelet Transform (WT) and Tunable Q-Factor Wavelet Transform (TQWT) features

are used to extract subtle abnormalities present in the production of speech and in the vibration of the vocal folds for speech-based PD detection [17–19]. These techniques are good at representing articulation patterns, frequency variations, and signal characteristics of PD.

3.2 Data Preprocessing

Data preprocessing is a crucial step for building robust models, particularly when handling high-dimensional features. Since the dataset contains both categorical and numerical variables, appropriate preprocessing techniques, including encoding, class balancing, and feature scaling, are applied to improve model generalization and reduce biased learning. As illustrated in Fig. 1, a preprocessing pipeline is employed before classification.

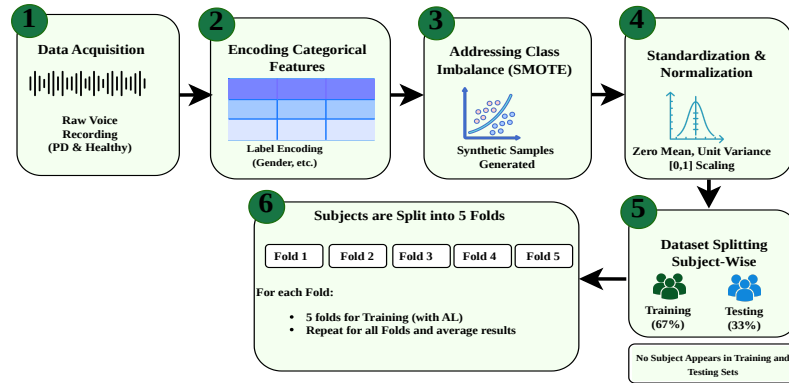


Fig. 1. Overall workflow demonstrating the whole process of the study.

- **Encoding Categorical Features:** Binary categorical variables, such as gender, are converted into numerical form using label encoding [20], enabling algorithms to handle them effectively.
- **Handling Class Imbalance:** To mitigate class imbalance and reduce model bias, the Synthetic Minority Over-sampling Technique (SMOTE) [21] is applied exclusively to the training set within each subject-wise CV fold. SMOTE is not applied to the test sets, ensuring that their original class distributions are preserved and preventing information leakage. At each ALL round, SMOTE is re-applied only to the currently labeled training subset.
- **Standardization and Normalization:** Balancing the delivery of a standard and a normal product. Delivering a standard and a normal product. Each of the features is standardized to a zero mean and unit variance and made normalized to prevent domination of a feature in the training process process [22].

- **Dataset Splitting:** To prevent information leakage, the dataset is partitioned at the subject level, ensuring that recordings from the same participant appear exclusively in either the training set or the test set. A 67/33 subject-wise split is used, with 67% of subjects allocated for training and 33% held out for test set. The test set remains completely isolated from AL, model training, and CV, which are conducted only within the 67% training set.

The 67/33 split and five-fold cross-validation serve different purposes. Five-fold subject wise CV is performed exclusively within the 67% training set. The results from the five folds are averaged for performance estimation, while the independent 33% test set remains reserved for unbiased final evaluation.

3.3 Proposed System Architecture of Cleanlab-Guided Uncertainty-based Active Learning Framework

The proposed architecture presents an uncertainty-driven AL framework for efficient PD classification using speech data while minimizing annotation cost. The complete workflow of the proposed method is described in Algorithm. 1, where the step-by-step procedure of the learning process is clearly outlined. Initially, the raw voice dataset is divided into subject-wise training and testing sets to prevent data leakage and ensure robust generalization. The training dataset is then processed using Cleanlab to identify and remove potential label issues, resulting in a cleaned training set. A small labeled seed data set is then chosen to train the first classification model, and the rest of the samples are put into an unlabeled data pool. In each AL iteration, the acoustic speech features extracted are fed into a feature encoder F to yield normalized latent representations and then passed to a discriminator-based classifier followed by a Softmax, which yield posterior probability distributions $P(Y|X)$. Model optimization is done using three losses: Binary Cross-Entropy $L(BCE)$, Entropy Loss $L(Ent)$, and Classification Error Loss $L(CE)$ to enhance the reliability of the probabilistic prediction. Uncertainty estimation mechanism is used to calculate AL scores for unlabeled samples and uncertainty sampling to determine the most informative samples for labeling in the acquisition function, which is implemented using a Cleanlab. These queried samples are simulated annotation oracle and iteratively incorporated into the labeled dataset for model retraining, progressively refining the decision boundary through a closed-loop learning process until convergence criteria are satisfied. Finally, the trained model is evaluated on a held-out test set to generate PD and healthy classification outputs, while SHAP and LIME are integrated to provide global and local interpretability, enabling the identification of important speech features contributing to PD detection. The overall workflow is illustrated in Fig. 2.

The methodology outlined in Algorithm 1 can be expressed through the following mathematical formulation. For each unlabeled sample, the AL score is computed using the prediction probabilities as defined in Eq. 1.

$$S_i = 1 - \max_{c \in C} P_U(x_i)_c \quad (1)$$

where $P_U(x_i)_c$ is the predicted probability for class c , a higher S_i indicates greater uncertainty of the model about x_i . A batch of B samples with the highest uncertainty is selected for labeling as illustrated in Eq. 2.

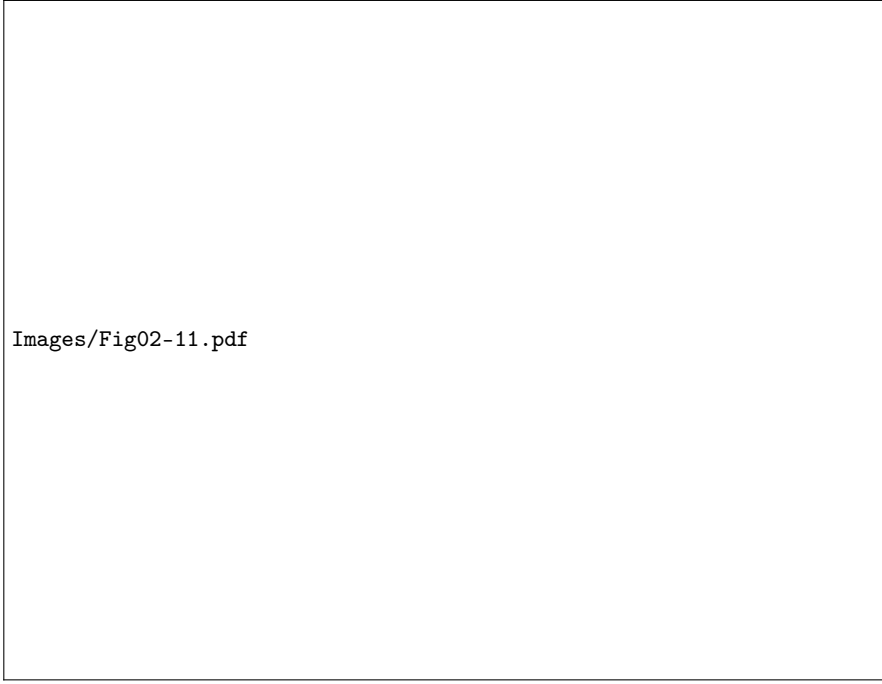


Fig. 2. Pipeline of Cleanlab-Guided Uncertainty-Based Active Learning Framework for PD Classification.

$$I = \arg \max_{I \subseteq U, |I|=B} \sum_{i \in I} S_i \quad (2)$$

4 Results and Discussion

This section provides a comprehensive overview of the experimental results, covering the experimental protocol and evaluation setup, improvement across rounds, a comprehensive ablation study, and the interpretability analysis with the help of SHAP and LIME.

4.1 Experimental Setup

The experiments are conducted in Python in a PyCharm integrated development environment (IDE) in a separate virtual environment (venv) to enable reproducibility and dependency management. All the required libraries and structures are installed and up-to-date in this isolated environment. To, make it clear, the study describes two model stages as follows:

Initial Model: Trained using only the initial small labeled subset before applying AL iterations.

Algorithm 1 PROPOSED UNCERTAINTY-BASED ACTIVE LEARNING ALGORITHM

```
1: Initialize labeled set  $\mathcal{L}$  and unlabeled set  $\mathcal{U}$  using  $\text{GETLabeled}(\mathcal{D}_{\text{train}}, k)$ 
2: for  $t \leftarrow 1$  to  $T$  do
3:    $M \leftarrow \text{TRAINMODEL}(\mathcal{L})$ 
4:    $P_{\mathcal{U}} \leftarrow M(\mathcal{U})$ 
5:    $S \leftarrow \text{COMPUTEACTIVELEARNINGScores}(\mathcal{L}, P_{\mathcal{U}})$ 
6:    $I \leftarrow \text{SELECTHIGHESTScores}(S, B)$ 
7:    $\mathcal{L} \leftarrow \mathcal{L} \cup \mathcal{U}[I]$ 
8:    $\mathcal{U} \leftarrow \mathcal{U} \setminus \mathcal{U}[I]$ 
9: end for
```

Final Model: Obtained after completing all AL rounds, where uncertainty-selected samples are added iteratively, and the model is retrained to achieve optimized performance. Table. 1 summarizes the main experimental settings used in the AL framework.

Table 1. Hyperparameters settings of proposed model

Component	Experimental Setting
Dataset partition	67% training set, 33% test set
Labelled seed	8 samples per class
Unlabelled pool	Remaining samples in the training
Query batch size	100 samples per round
Active-learning rounds	20
Repeated seeds	Multiple random seeds
Cross-validation	Subject-wise 5-fold CV
Stopping rule	Fixed 20 rounds

4.2 Performance Improvement Across Rounds

This subsection assesses the effectiveness of the proposed AL strategy by examining the round-wise performance progression on the full feature set.

The proposed method demonstrates a consistent improvement in classification performance as the AL rounds progress. As shown in Fig. 3, there is progressive enhancement of the classification accuracy in relation to the round. The model starts with an accuracy value of 74.9% in Round 1 and a highest value of 94.0% in Round, 8 and the same is upheld in subsequent rounds. The dark area is the standard deviation of variability over the runs, which decreases with the round. Fig. 4 indicates that the F1-score has increased to 94.0% in Round 8, but it has not improved since then. On the same note, the increase in MCC as demonstrated in the same Fig. 5 shows that the MCC is increasing steadily to get to 88.1% in Round 8, which is a reliable and consistent classification performance. Fig. 6 depicts the improvement in recall with communication rounds. In Round 1, the model starts with a low recall of 74.2% ,and it progressively reaches a peak of 93.9% in Round 8, and then the model is maintained. Fig. 7 of the performance shows the trend of the RMSE and it starts with a value of 0.496 and then decreases to 0.241 and thereafter it

does not change but shows similar values hence more accurate predictions and thus more accurate predictions. Fig. 8 MAE decreases drastically in Round 1 of 0.251 to Round 8 of 0.060 and then stabilizes until Round 20, indicating the convergence of the model.

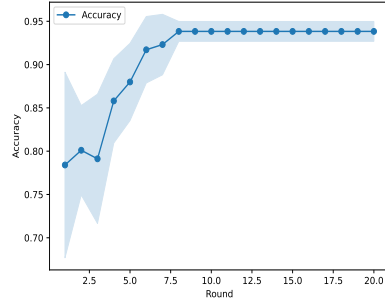


Fig. 3. Accuracy progression across rounds

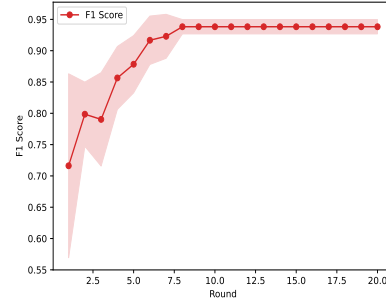


Fig. 4. F1-Score progression across rounds

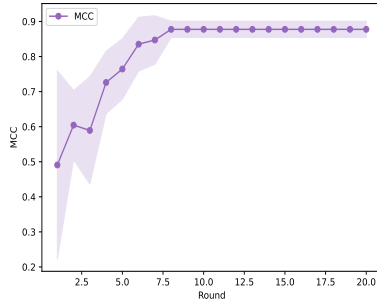


Fig. 5. MCC progression across rounds

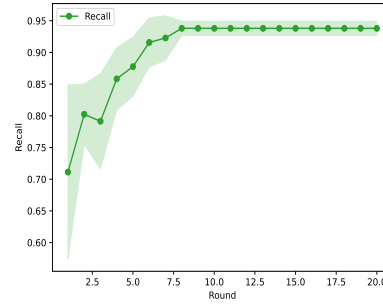


Fig. 6. Recall progression across rounds

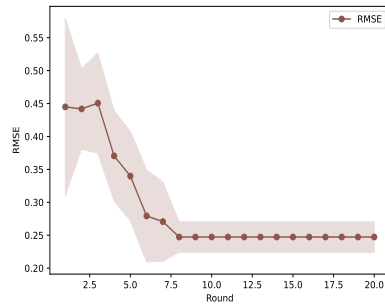


Fig. 7. RMSE progression across rounds

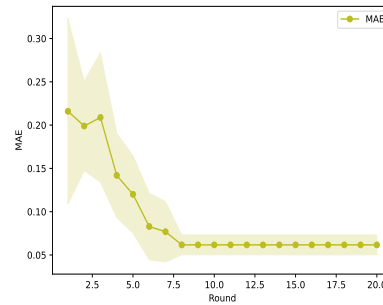


Fig. 8. MAE progression across rounds

4.3 Feature Group Ablation Study

The ablation analysis demonstrates clear and interpretable variations in performance across different feature groups. The ablation results in Table. 2 reveal clear and interpretable performance variations across individual feature groups. The most significant degradation occurs when TQWT features are removed, with MCC decreasing from 88.1% to 79.4%, and accuracy from 94% to 89.6%. The importance of TQWT in capturing the discriminative nonlinear time-frequency attributes that are required to classify PD is highlighted by this important loss. The removal of MFCC features also leads to a marked performance reduction of MCC from 88.1% to 83.1%, and accuracy from 94% to 92.3%. On the other hand, the performance is also slightly increased when time-frequency and vocal fold features lead to a slight performance improvement (MCC increasing to 89.4% and 89.3%, respectively). On the whole, it is evident that TQWT and MFCC are the most efficient sets of features in the specified framework, and the lack of either of them leads to a considerable decrease in the classification performance.

Table 2. Performance comparison under different excluded feature settings. The best three results are colored Green, Blue, and Red.

Excluded Features	Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	MCC (%)	RMSE	MAE
Baseline	Initial Model	0.862 $\pm$ 0.058	0.866 $\pm$ 0.057	0.861 $\pm$ 0.058	0.861 $\pm$ 0.058	0.727 $\pm$ 0.116	0.361 $\pm$ 0.087	0.138 $\pm$ 0.058
	Final Model	0.929 $\pm$ 0.018	0.932 $\pm$ 0.018	0.929 $\pm$ 0.019	0.929 $\pm$ 0.018	0.861 $\pm$ 0.037	0.264 $\pm$ 0.033	0.071 $\pm$ 0.018
MFCC	Initial Model	0.837 $\pm$ 0.054	0.854 $\pm$ 0.056	0.833 $\pm$ 0.069	0.826 $\pm$ 0.060	0.685 $\pm$ 0.117	0.398 $\pm$ 0.068	0.163 $\pm$ 0.054
	Final Model	0.923 $\pm$ 0.016	0.924 $\pm$ 0.016	0.922 $\pm$ 0.016	0.922 $\pm$ 0.016	0.831 $\pm$ 0.032	0.277 $\pm$ 0.030	0.077 $\pm$ 0.016
TQWT	Initial Model	0.733 $\pm$ 0.030	0.719 $\pm$ 0.034	0.713 $\pm$ 0.034	0.715 $\pm$ 0.034	0.431 $\pm$ 0.068	0.516 $\pm$ 0.030	0.267 $\pm$ 0.030
	Final Model	0.896 $\pm$ 0.037	0.898 $\pm$ 0.036	0.896 $\pm$ 0.037	0.896 $\pm$ 0.037	0.794 $\pm$ 0.073	0.315 $\pm$ 0.068	0.104 $\pm$ 0.037
Wavelet	Initial Model	0.784 $\pm$ 0.063	0.801 $\pm$ 0.076	0.784 $\pm$ 0.064	0.781 $\pm$ 0.062	0.585 $\pm$ 0.139	0.460 $\pm$ 0.070	0.216 $\pm$ 0.063
	Final Model	0.932 $\pm$ 0.031	0.933 $\pm$ 0.029	0.931 $\pm$ 0.031	0.932 $\pm$ 0.031	0.865 $\pm$ 0.060	0.254 $\pm$ 0.063	0.068 $\pm$ 0.031
Vocal Fold	Initial Model	0.776 $\pm$ 0.063	0.790 $\pm$ 0.067	0.765 $\pm$ 0.067	0.764 $\pm$ 0.072	0.553 $\pm$ 0.130	0.460 $\pm$ 0.069	0.224 $\pm$ 0.063
	Final Model	0.946 $\pm$ 0.025	0.947 $\pm$ 0.025	0.946 $\pm$ 0.025	0.946 $\pm$ 0.025	0.893 $\pm$ 0.049	0.225 $\pm$ 0.056	0.054 $\pm$ 0.025
Time Frequency	Initial Model	0.801 $\pm$ 0.053	0.837 $\pm$ 0.038	0.791 $\pm$ 0.061	0.789 $\pm$ 0.064	0.625 $\pm$ 0.096	0.442 $\pm$ 0.060	0.199 $\pm$ 0.053
	Final Model	0.946 $\pm$ 0.018	0.948 $\pm$ 0.017	0.946 $\pm$ 0.018	0.946 $\pm$ 0.018	0.894 $\pm$ 0.035	0.229 $\pm$ 0.037	0.054 $\pm$ 0.018
Full Dataset	Initial Model	0.749 $\pm$ 0.066	0.761 $\pm$ 0.064	0.737 $\pm$ 0.075	0.737 $\pm$ 0.075	0.501 $\pm$ 0.135	0.496 $\pm$ 0.073	0.251 $\pm$ 0.066
	Final Model	0.940 $\pm$ 0.023	0.941 $\pm$ 0.023	0.939 $\pm$ 0.023	0.940 $\pm$ 0.023	0.881 $\pm$ 0.046	0.241 $\pm$ 0.047	0.060 $\pm$ 0.023

4.4 Comparative Evaluation of Active Learning Acquisition Strategies

Table. 3 compares five AL acquisition strategies. Cleanlab-Guided Sampling achieves the best overall performance, with an accuracy of 0.940 ± 0.023 , precision of 0.940 ± 0.023 , and MCC of 0.881 ± 0.046 . Random Sampling achieves the second-best performance, while Entropy, Margin, and Least-Confidence Sampling show similar and comparatively lower results. Overall, the results demonstrate the effectiveness of the proposed Cleanlab-Guided Sampling strategy.

4.5 Comparison with Existing SSL models

Following the experimental evaluation of the proposed framework, its effectiveness is further assessed by benchmarking against several state-of-the-art SSL models on the same

Table 3. Performance comparison of different Active Learning acquisition strategies.

Method	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	MCC (%)
Random Sampling	0.863 ± 0.021	0.871 ± 0.011	0.864 ± 0.020	0.862 ± 0.022	0.734 ± 0.030
Entropy Sampling	0.848 ± 0.039	0.861 ± 0.035	0.849 ± 0.038	0.847 ± 0.040	0.710 ± 0.073
Margin Sampling	0.848 ± 0.039	0.861 ± 0.035	0.849 ± 0.038	0.847 ± 0.040	0.710 ± 0.073
Least-Confidence Sampling	0.848 ± 0.039	0.861 ± 0.035	0.849 ± 0.038	0.847 ± 0.040	0.710 ± 0.073
Proposed Model	0.940 ± 0.023	0.940 ± 0.023	0.930 ± 0.023	0.930 ± 0.023	0.881 ± 0.046

datasets [24]. The comparison included Rasco [25] and RelRasco [26], which employ ensemble strategies combined with subspace selection, as well as CoTrainingByCommittee [27] and CoTraining [28], both of which expand the labeled set using multiple classifiers or views. Additionally, Setred [29] incorporates a rejection mechanism to filter noisy predictions, while DeTriTraining [30] leverages consensus among multiple classifiers to assign labels to unlabeled data. Other approaches, including CoForest [31], TriTraining [32], DemocraticCoLearning [33], and SelfTraining [34], utilize ensemble agreement rules, diverse classifier integration, or iterative high-confidence retraining to enhance labeling reliability.

Table 4. Comparison of the proposed method with baseline SSL methods.

Model	Accuracy (%)	F1 -Score (%)	ROC-AUC (%)	MCC (%)	Train Time (s)
Rasco [25]	0.8440 ± 0.0349	0.8438 ± 0.0350	0.9179 ± 0.0281	0.6887 ± 0.0690	16.2906 ± 0.6731
RelRasco [26]	0.8413 ± 0.0483	0.8411 ± 0.0484	0.9152 ± 0.0360	0.6843 ± 0.0963	17.3634 ± 1.1337
CoTrainingByCommittee [27]	0.7704 ± 0.0238	0.7693 ± 0.0237	0.8493 ± 0.0449	0.5461 ± 0.0487	27.1915 ± 2.6403
CoTraining [28]	0.7429 ± 0.0189	0.7422 ± 0.0189	0.7429 ± 0.0189	0.4888 ± 0.0382	6.6953 ± 0.3016
Setred [29]	0.7420 ± 0.0322	0.7411 ± 0.0326	0.7419 ± 0.0323	0.4872 ± 0.0639	1.9757 ± 0.3437
DeTriTraining [30]	0.6924 ± 0.0239	0.6921 ± 0.0241	0.6924 ± 0.0240	0.3856 ± 0.0477	220.0038 ± 31.8402
CoForest [31]	0.6862 ± 0.0369	0.6853 ± 0.0372	0.7071 ± 0.0347	0.3745 ± 0.0735	9.9148 ± 1.3377
TriTraining [32]	0.7660 ± 0.0201	0.7635 ± 0.0214	0.8643 ± 0.0221	0.5426 ± 0.0388	0.2983 ± 0.0367
Democratic Co-Learning [33]	0.7544 ± 0.0469	0.7500 ± 0.0499	0.8214 ± 0.0241	0.5238 ± 0.0846	8.7140 ± 1.3751
Self-Training [34]	0.5000 ± 0.0000	0.5000 ± 0.0000	0.5000 ± 0.0000	0.6426 ± 0.0388	8.9148 ± 1.3378
Proposed Model	0.941 ± 0.023	0.940 ± 0.023	0.7816 ± 0.0315	0.881 ± 0.046	28.2966 ± 0.5478

Table. 4 indicates the comparison of the proposed AL model with the baselines of SSL using 20% of labeled data. TriTraining illustrates the shortest training time ($0.2983 \pm 0.0367s$), but the cost is a significantly reduced accuracy, F1-Score, MCC, and ROC-AUC than the suggested model. The best base line strategies that have the closest accuracy to the proposed model are Rasco and RelRasco, which are 84.4% (± 0.0349) and 84.13% (± 0.0483), respectively. However, a significant relative improvement of about 12.3% over Rasco and 12.6% over RelRasco is provided by the proposed framework.

4.6 SHAP: Global Interpretability

SHAP is a model-agnostic interpretability technique that explains ML predictions by assigning Shapley values to features based on their contributions to model outputs. In this study, SHAP is employed to provide global model interpretability by identifying the most influential features affecting prediction performance. The Shapley value for a feature b is computed as defined in Eq. 3.

$$\phi_b = \sum_{S \subseteq N \setminus \{b\}} \frac{|S|! (|N| - |S| - 1)!}{|N|!} [v(S \cup \{b\}) - v(S)] \quad (3)$$

where ϕ_j is the Shapley value associated with feature b , S is a feature subset excluding b , N represents the set of all features, and $v(S)$ denotes the model output corresponding to the feature set S .

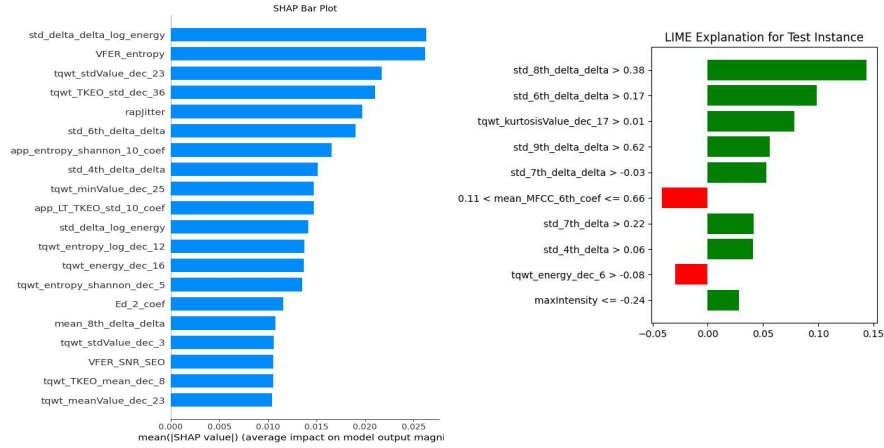


Fig. 9. SHAP-based feature importance. **Fig. 10.** LIME-based local explanation.

The global feature importance by mean absolute SHAP values is shown in Fig. 9. Parameters `std_delta_delta_log_energy`, `VFER_entropy`, `tqwt_stdValue_dec_23`, and `tqwt_TKEO_std_dec_36` have the greatest contribution to the model output, which shows that they have the greatest effect on the classification process. On the other hand, `VFER_SNR_SEO`, `tqwt_TKEO_mean_dec_8`, and `tqwt_meanValue_dec_23` contribute less to the model predictions.

4.7 LIME: Local Interpretability

LIME provides insights into complex models by generating local explanations for individual predictions. It works by perturbing an instance’s input features and observing the resulting changes in model output. To approximate the behavior of the complex model locally, LIME learns a surrogate model $\hat{h}$ by minimizing the objective function defined in Eq. 4.

$$\hat{h} = \arg \min_{h \in H} L(f, h, \pi_x) + \Omega(h) \quad (4)$$

where f is the original complex model, h is the interpretable surrogate, and $\Omega(h)$ penalizes the complexity of h to ensure interpretability. Fig. 10 presents the LIME-based local explanation for a single test instance, illustrating how individual features influence the model’s prediction. Green bars indicate features that positively contribute toward the predicted class, while red bars represent features that negatively impact the prediction. Features such as `std_8th_delta_delta`, `std_6th_delta`, and `tqwt_kurtosisValue_dec_17` show the strongest positive contributions, whereas features exert a negative influence including `mean_MFCC_6th_coef`, and `tqwt_entropy_dec_6`.

4.8 Comparison with Existing Literature

Recent studies have investigated PD classification, and Table. 5 summarizes the performance of several existing methods. Our model yields the highest accuracy, which is 94.0%, and MCC, 88.1%, which is higher than the other existing methods. This evidences that the suggested framework has better diagnostic performance and optimally lowers labeling needs. While some previous studies, such as Yue et al. [7] reported accuracy ranging from 44.3% with annotated samples of 25%. Bao et al. [9] gained an accuracy of 83.4% and an MCC of 67.5%. Although Adnan et al. [11] found a superior MCC, they experienced lower F1-Scores of 77.93% and accuracy of 85.7%, indicating that the proposed framework can provide a more consistent and accurate performance in classification.

Table 5. Comparison with existing literature

Author	Accuracy (%)	F1-Score (%)	MCC (%)
Yue et al. [7]	44.3%–90%	-	-
Bao et al. [9]	83.80%	-	67.50%
Adnan et al. [11]	85.70%	77.93%	97%
Sakar et al. [10]	89%	84%	59%
Proposed Model	94%	94%	88.1%

5 Discussion

The experimental findings clearly demonstrate the effectiveness of the proposed framework in achieving high diagnostic performance with minimal labeled data. In comparison

to the traditional methods of supervision and semi-supervision, the suggested method is strategic in choosing the most informative samples, cutting annotation needs by a considerable margin, and not compromising the classification accuracy. One of the main observations made based on the results is the fast improvement in the performance throughout the initial AL rounds. Significant improvements are observed in the first 8 iterations with an accuracy of 94.0% and a MCC of 88.1%. The feature contributions mentioned in the ablation study show that the removal of TQWT and MFCC features result in a significant drop in performance, thus proving that they have a great contribution to the PD acoustic pattern. SHAP analysis shows that energy- and entropy-based features are most influential in global predictions, while LIME provides instance-level explanations supporting feature-wise decision understanding. Overall, the proposed uncertainty-based AL framework with XAI offers an effective, scalable, and cost-efficient approach for PD diagnosis under limited labeled data.

6 Conclusion

This study presents a Cleanlab-Guided Uncertainty-Based Active Learning framework for efficient PD classification using high-dimensional voice-based acoustic features. The proposed approach addresses the problem of the scarcity of labeled data by biased sampling of the most informative samples of an unlabeled pool, thereby reducing annotation cost while maintaining strong diagnostic performance. The experimental findings indicate that the model attains an accuracy of 94.0%, an F1-Score of 94.0%, and an MCC of 88.1% and it reaches a high classification performance within eight AL rounds, implying that high classification accuracy can be achieved with a relatively small number of labeled samples. A comprehensive ablation study is conducted to assess the contribution of individual components within the framework, confirming the effectiveness of the proposed framework. Furthermore, a comparative analysis with existing SSL models highlights the robustness and superior performance of the proposed method. In addition, SHAP and LIME integration offer global and local interpretability, showing the most significant features. Overall, the findings suggest that combining uncertainty-driven active learning with explainable AI offers a scalable, cost-effective, and reliable framework for voice-based PD diagnosis. The future work will be done on multilingual validation and a bigger clinical evaluation.

Code and Data Availability

The dataset used in this study is available from the UCI Machine Learning Repository. The code supporting the findings of this study are available from the corresponding author upon reasonable request.

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